# VENTILATION-FREE DAYS TO ESTIMATE THE BENEFIT OF CORTICOTHERAPY FOR THE TREATMENT OF COMMUNITY-ACQUIRED PNEUMONIA IN HOSPITALIZED PATIENTS

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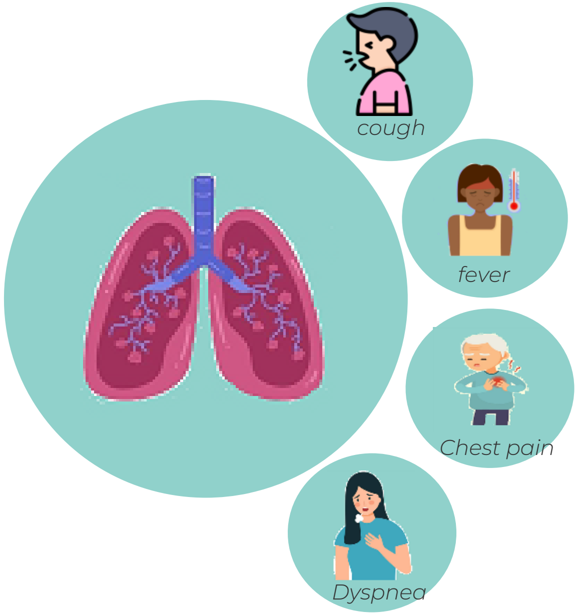
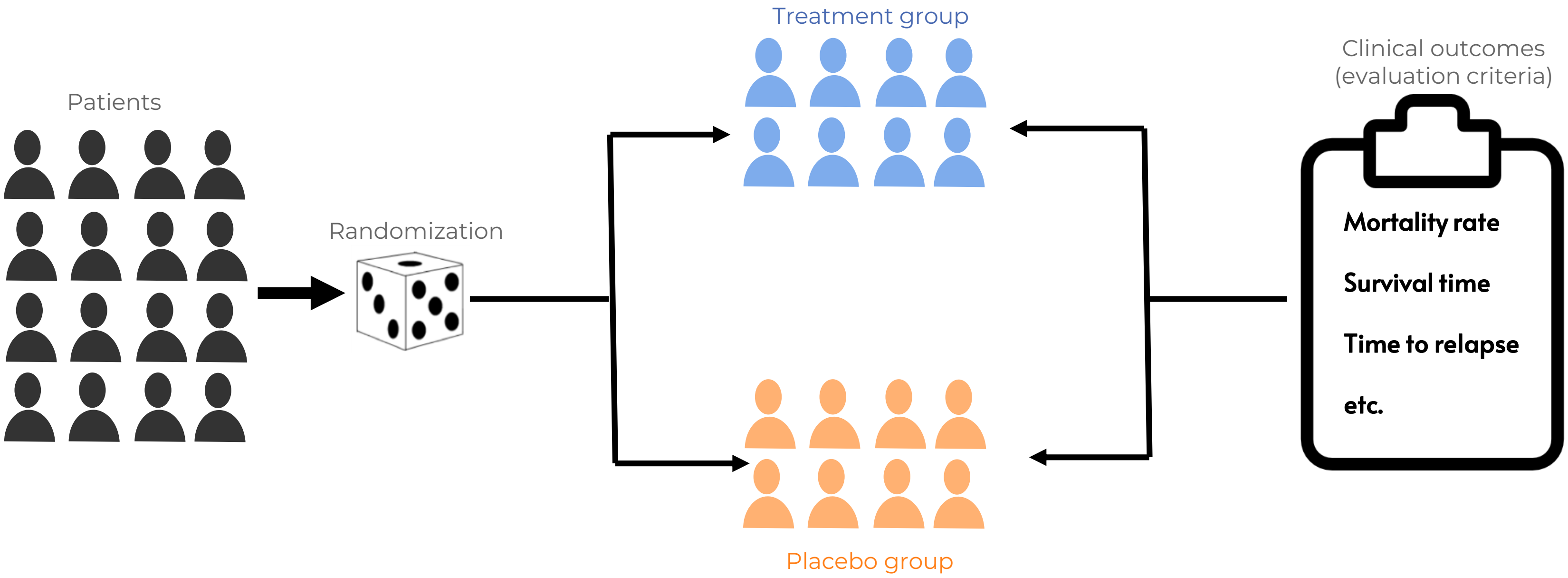
# INTRODUCTION



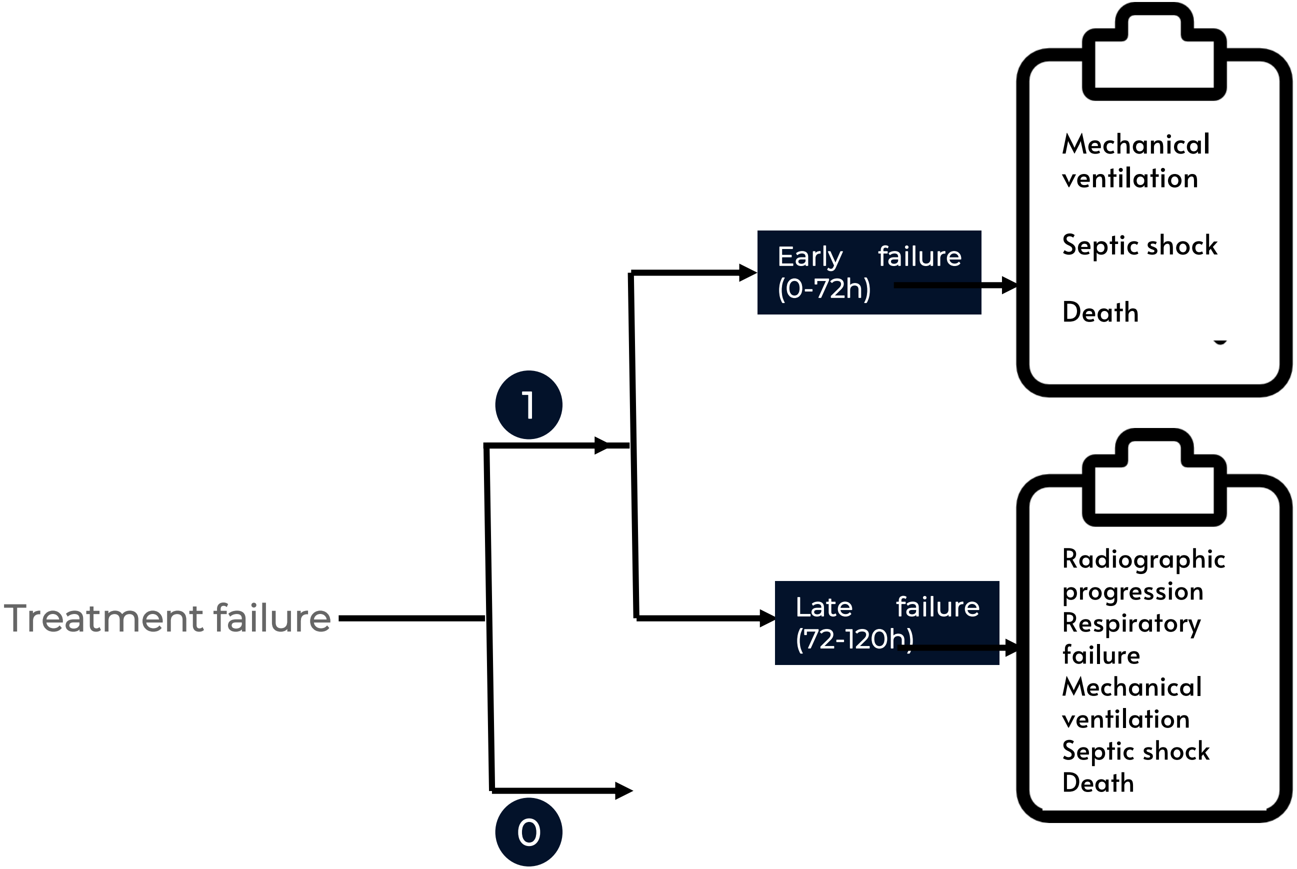

- Community-acquired pneumonia (CAP) is an acute disease, caused by an infection of the lung, acquired outside of a hospital setting.

- Promising data have been on the administration of corticosteroids as coadjuvant treatment in hospitalized patients with CAP, but their use is still controversial

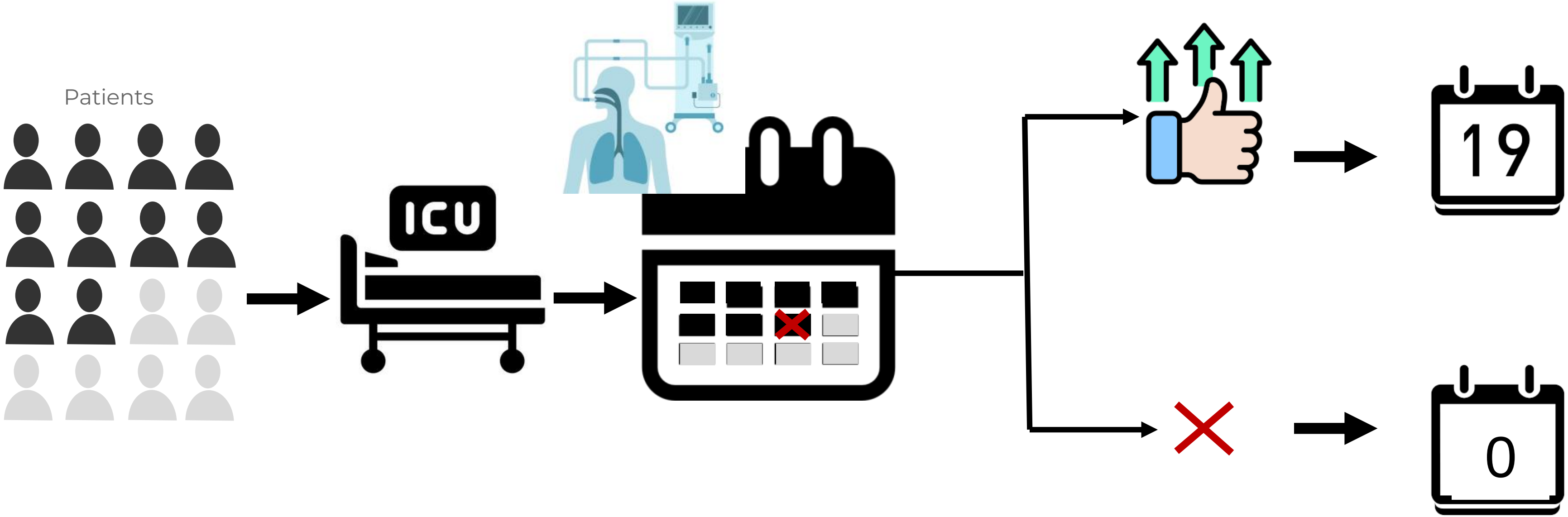
## Randomized Clinical trials



# INTRODUCTION



## Ventilation-free days (VFDs)



# INTRODUCTION

**Main objective:** reanalyze a clinical trial in order to evaluate the effect of corticosteroids on clinical outcomes in patients with severe CAP, with particular attention to VFDs, a novel outcome introduced in the present analysis

Research

Original Investigation | CARING FOR THE CRITICALLY ILL PATIENT

## Effect of Corticosteroids on Treatment Failure Among Hospitalized Patients With Severe Community-Acquired Pneumonia and High Inflammatory Response

### A Randomized Clinical Trial

Antoni Torres, MD, PhD; Oriol Sibila, MD, PhD; Miquel Ferrer, MD, PhD; Eva Polverino, MD, PhD; Rosario Menendez, MD, PhD; Josep Mensa, MD, PhD; Albert Gabarrús, MSc; Jacobo Sellarés, MD, PhD; Marcos I. Restrepo, MD, MSc; Antonio Anzueto, MD, PhD; Michael S. Niederman, MD; Carles Agustí, MD, PhD

# INTRODUCTION

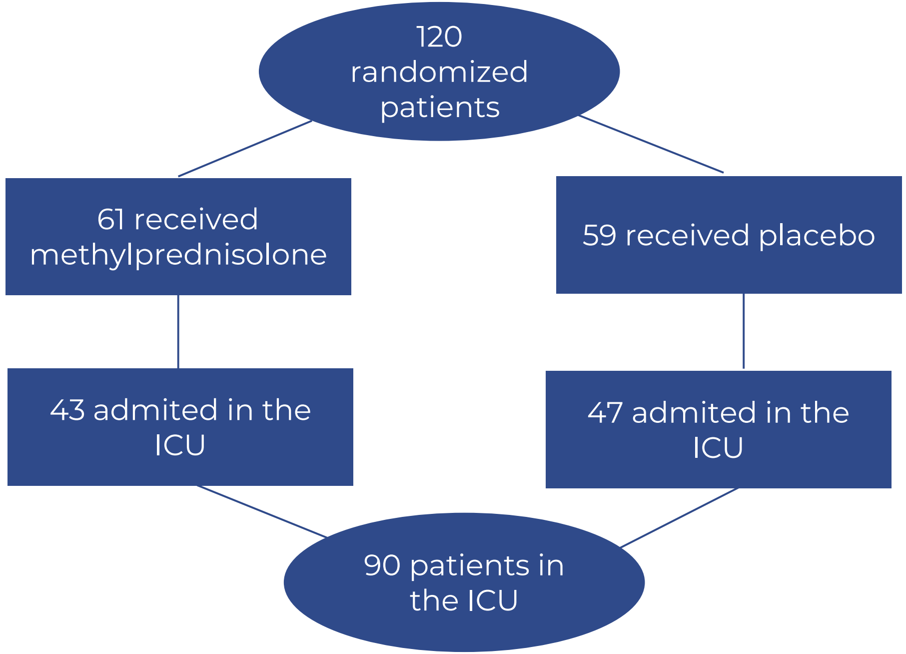
**Main objective:** reanalyze a clinical trial in order to evaluate the effect of corticosteroids on clinical outcomes in patients with severe CAP, with particular attention to VFDs, a novel outcome introduced in the present analysis

01 Assess the effect of corticosteroids, precisely methylprednisolone, on clinical outcomes.

02 Assess the effect of corticosteroids on VFDs, including an implicit methodological comparison between frequentist and Bayesian methods.

## Study design

Double-blind with patients enrolled and followed up across three Spanish hospitals from June 2004 through February 2012.



# INTRODUCTION

## Study design

### Primary outcomes

**Treatment failure* 20% (n=24)**

| | | Global (n=24/120) | Treatment (n=7/61) | Placebo (n=17/59) |
|---|---|---|---|---|
| Early failure (0-72h) | | 10% (n=12) | 9.8% (n=6) | 10.2% (n=6) |
| | Mechanical ventilation | **7.5% (n=9)** | 6.6% (n=4) | 8.5% (n=5) |
| | Septic shock | 4.2% (n=5) | 3.3% (n=2) | 5.1% (n=3) |
| | Death | 3.3% (n=4) | 3.3% (n=2) | 3.4% (n=2) |
| Late Failure* (72-120h) | | 12.5% (n=15) | 1.6% (n=1) | 23.7% (n=14) |
| | Radiographic progression* | **8.3% (n=10)** | 1.6% (n=1) | 15.3% (n=9) |
| | Respiratory failure | 5% (n=6) | 1.6% (n=1) | 8.5% (n=5) |
| | Mechanical ventilation | 4.2% (n=5) | 1.6% (n=1) | 6.8% (n=4) |
| | Septic shock | 3.3% (n=3) | 0 | 6.8% (n=4) |
| | Death | 0 | 0 | 0 |

# 01

## Corticosteroids on ventilation-free days

# Corticosteroids on ventilation-free days

## Methodology

### Model choice

- Count data
- Pearson chi-square overdispersion test**
- Comparison of proportion of observed zeros (12.2%) vs negative binomial zeros (0.41%)

$$\text{VFDs} = \begin{cases} 0, \text{ if death or stay >28 days} \\ 28 - \text{length of stay in ICU otherwise} \end{cases}$$

X: treatment group X ∈ {0, 1}
Y: VFDs , Y ∈ ⟦0, 28⟧
Z: death, Z ∈ {0, 1}.

n=number of patients in ICU (90)

**Mutia Sari and Open Darnius. Zero-inflated poisson regression testing in handling overdispersionon poisson regression. Journal of Mathematics Education and Application (JMEA), 2(2):96–107,2023.

$$\begin{cases} Y \mid Z = 0, X = x \sim \text{NB}(\mu_x, \theta),\ x \in \{0, 1\},\ \theta > 0 \\ Y \mid Z = 1, X = x \sim \text{B}(\pi(x)) \end{cases}$$

- $\mu_x = e^{\beta 0 + \beta 1 x}$, the average number of VFDs, dependent on the treatment group
- $\pi(x) = P(Z = 1 \mid X = x) = \frac{e^{\alpha 0 + \alpha 1 x}}{1 + e^{\alpha 0 + \alpha 1 x}}$ the probability of dying considering the treatment group,

Zero-inflated negative binomial model:

$$f_{Y|X,Z}(y \mid x, z) = \begin{cases} \left[\pi(x) + (1 - \pi(x)) \left(\frac{\theta}{\theta + \mu_x}\right)^{\theta}\right] \cdot \mathbb{1}_{\{\mu_x > 0, \theta > 0\}} \cdot \mathbb{1}_{x \in \{0,1\}}, & \text{if } y=0 \\ (1 - \pi(x)) \frac{\Gamma(y+\theta)}{y!\,\Gamma(\theta)} \left(\frac{\theta}{\theta + \mu_x}\right)^{\theta} \left(\frac{\mu_x}{\theta + \mu_x}\right)^{y} \cdot \mathbb{1}_{\mathbb{N}}(y) \cdot \mathbb{1}_{\{\mu_x > 0, \theta > 0\}} \cdot \mathbb{1}_{x \in \{0,1\}}, & \text{if } y>0 \end{cases}$$

# Corticosteroids on ventilation-free days

## Methodology

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***Mutia Sari and Open Darnius. Zero-inflated poisson regression testing in handling overdispersionon poisson regression. Journal of Mathematics Education and Application (JMEA), 2(2):96–107,2023.*

## Bayesian

Prior densities:

$$\gamma(\alpha,\beta) = \frac{exp(-\frac{1}{2}((\alpha,\beta)-\mu)^{\top}\Sigma^{-1}((\alpha,\beta)-\mu))}{2\pi^2|\Sigma|^{\frac{1}{2}}}$$

$$\phi(\theta) = \theta^{a-1}\frac{b^a e^{-b\theta}}{\Gamma a}$$

with:

- $p(x) = \frac{\theta}{\theta + \mu_x}$
- $\theta \sim Gamma\ (0.05, 0{,}01)$ ⟶ 95%CI=]0, 56.7]
- $\alpha_0, \alpha_1 \sim N(0{,}10)$ ⟶ 95%CI=]0, 1[

} Non informative

- $\beta_0 \sim N(2.3{,}0.3)$ ⟶ 95%CI=]3.4, 29.18[
- $\beta_1 \sim N(1.1{,}0.3)$ ⟶ 95%CI=]1.02, 8.76[

} Optimistic

Likelihood: $f(y \mid \alpha,\beta,\theta) = \pi(x)\mathbf{1}_{y=0} + (1-\pi(x))\binom{y+\theta-1}{y}p(x)^{\theta}(1-p(x))^{y}$

Posterior distribution:

$$\boxed{\psi(\alpha,\beta,\theta \mid y) \approx \theta^{a-1}\frac{b^a e^{-b\theta}}{\Gamma(a)}\frac{exp(-\frac{1}{2}((\alpha,\beta)-\mu)^{\top}\Sigma^{-1}((\alpha,\beta)-\mu))}{2\pi^2|\Sigma|^{\frac{1}{2}}}\left[\pi(x)\mathbf{1}_{y=0} + (1-\pi(x))\binom{y+\theta-1}{y}p(x)^{\theta}(1-p(x))^{y}\right]}$$

Gibbs sampling method using JAGS on R

# Corticosteroids on ventilation-free days

## Results

### Frequentist

| | Coefficients | Std. Error | 95% CI | $p$ |
|---|---|---|---|---|
| $\alpha_0$* | -3.78 | 0.07 | [-3.780423, -3.78042302] | 0.00 |
| $\alpha_1$* | -0.90 | 0.11 | [-1.21, -0.45] | 0.00 |
| $\beta_0$* | 3.17 | 0.01 | [3.174991 3.1749906] | 0.00 |
| $\beta_1$ | 0.02 | 0.01 | [0.00, 0.03] | 0.07 |

*Significant at 5% level*

| | |
|---|---|
| Estimated mortality rate | 0.0223<br>0.0092 |
| Estimated average number of days | 23.93<br>24.37 |

- We fail to find significant effect of treatment on number of VFDs.
- The odds of death under the treatment are 59% lower than the odds under the placebo (OR =0.41 [0.29 , 0.64] ).
- The model underestimates the probability of death, which is 6.67% for ICU patients in the placebo group (n=7), and 2.27% for ICU patients in the treatment group (n=3).

### Bayesian

Number of chains : 3
Sample size per chain : 1666
Thinning : 15

| | Coefficients | Std. Error | 95% credible interval |
|---|---|---|---|
| $\alpha_0$ | -0.83 | 0.96 | [-3.74, 1.06] |
| $\alpha_1$ | -0.94 | 0.71 | [-2.42, 0.38] |
| $\beta_0$ | 3.12 | 0.08 | [2.97 3.28] |
| $\beta_1$ | 0.04 | 0.05 | [-0.06, 0.14] |
| $\theta$ | 165.31 | 95.79 | [51.57, 410.11] |

$\mathbb{P}(\alpha_1 < \alpha_0) = 0.52$

$\mathbb{P}(\beta_1 > 0) = 0.78$

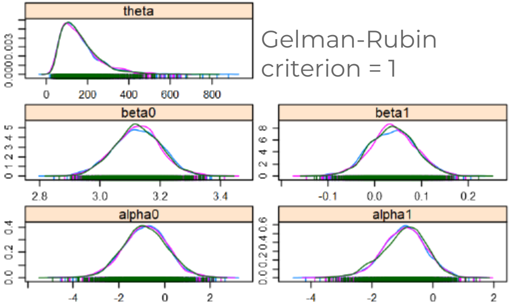
| | |
|---|---|
| Estimated mortality rate | 0.30<br>0.14 |
| Estimated average number of days | 22.75<br>23.64 |

- **The treatment leads to an increase of 4% [-0.06%, 15%] of the number of VFDs,** going from an average of 22.75 days for patients in the placebo group, to 23.64 days in the treatment group.
- The estimations are close to the true values with a true average of 23.34 days for ICU patients in the placebo group, and 23.78 days for those in the treatment group
- The probability that $\alpha1 < \alpha0$ is 52%, there might be a very mild to no effect of the methylprednisolone on mortality rate.

# Corticosteroids on ventilation-free days

## Bayesian model diagnostic

- Gelman-Rubin criterion
- Within-chain autocorrelation
- Effective sample size (ESS)
- Comparison true mean vs predicted




Gelman-Rubin criterion = 1

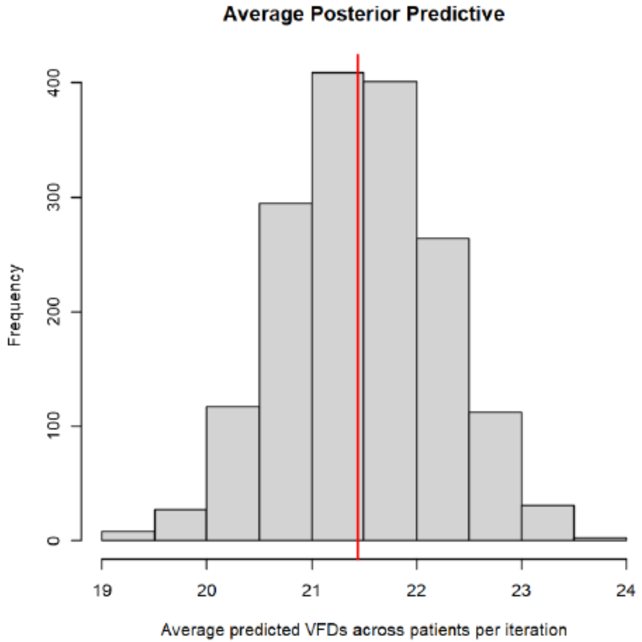
*Overlaid density curves of each chain for each parameter*

| | $\alpha_0$ | $\alpha_1$ | $\beta_0$ | $\beta_1$ | $\theta$ |
|---|---|---|---|---|---|
| Lag 0 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| Lag 15 | 0.28 | 0.29 | 0.36 | 0.36 | -0.00 |
| Lag 75 | -0.03 | -0.02 | 0.01 | 0.01 | 0.02 |
| Lag 150 | 0.01 | 0.01 | -0.02 | -0.03 | -0.00 |
| Lag 750 | 0.01 | 0.01 | 0.02 | 0.01 | 0.00 |

*Within-chain autocorrelation*

| | $\alpha_0$ | $\alpha_1$ | $\beta_0$ | $\beta_1$ | $\theta$ |
|---|---|---|---|---|---|
| ESS | 2732 | 2641 | 2338 | 2317 | 4998 |

*ESS for parameters*

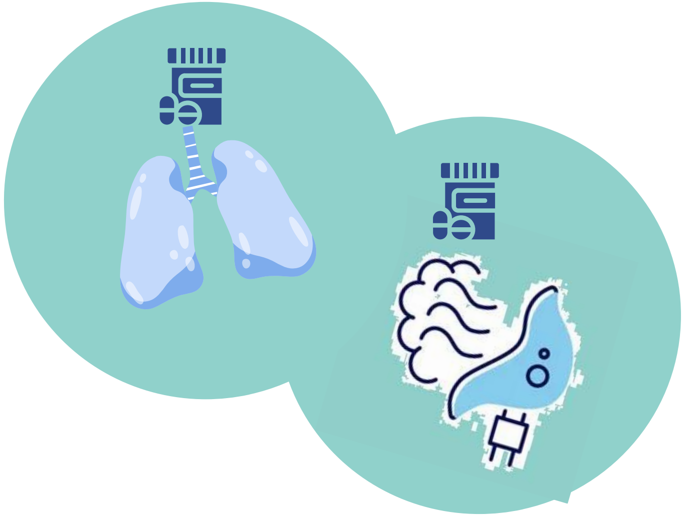



*True mean (red line) vs Predicted mean of VFDs*

The observed mean of VFDs for ICU patients lies inside the central part of the simulated distribution (i.e region where the model predicts values the most), so the model is likely well-fitted.

# 02

## To go further



# To go further

**Goal:**

Establish a methodological framework to evaluate the potential impact of baseline characteristics on the effect of treatment on treatment failure and VFDs

## Goals

The discrepancy in the results could be explained by the interaction with baseline characteristics

$$\text{comorbidity} = \begin{cases} 1, & \text{if patient has a pre-existing comorbid condition*} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{severity} = \begin{cases} 1, & \text{if risk class} > 3^{**} \\ 0 & \text{otherwise} \end{cases}$$

** diabetes mellitus, Chronic pulmonary disease, congestive heart failure, cancer and ischemic heart disease.*

*** derived from Pneumonia Severity Score (PSI)*

# To go further

X: the random variable representing the treatment group (1 for treatment), $X \in \{0, 1\}$

Y: random variable representing the treatment failure (1 for failure), $Y \in \{0, 1\}$

- $W \in Rp$: a vector of covariates (terms susceptible to interact with the main covariate X)

## Goals

### Treatment failure

$Y = y \mid X = x, W = w \sim B(p(x, w))$, $x, y \in \{0, 1\}$, $w \in Rp$, $f(y \mid x, w) = p(x, w)^y (1 - p(x, w))^{1-y}$ with:

$$p(x,w) = \mathbb{P}(Y = 1 \mid X = x, W = w) = \frac{exp[\beta_0 + \beta_1 x + \sum_{j=1}^{p} \gamma_j w_j + \sum_{j=1}^{p} \nu_j (x \odot w_j)]}{1 + exp[\beta_0 + \beta_1 x + \sum_{j=1}^{p} \gamma_j w_j + \sum_{j=1}^{p} \nu_j (x \odot w_j)]} \cdot \mathbf{1}_{x,y \in \{0,1\}} \cdot \mathbf{1}_{w \in \mathbb{R}^p}$$

The likelihood for i.i.d sample (Yn, Xw, Wn) drawn from (Y,X,W):

$$L(\theta) = \prod_{i=1}^{n} \left[ \frac{(exp[\beta_0 + \beta_1 X_i + \sum_{j=1}^{p} \gamma_j W_{ij} + \sum_{j=1}^{p} \nu_j (X_i \odot W_{ij}]))^{Y_i}}{1 + exp(\beta_0 + \beta_1 X_i + \sum_{j=1}^{p} \gamma_j W_{ij} + \sum_{j=1}^{p} \nu_j (X_i \odot W_{ij}))} \right] \cdot \mathbf{1}_{X_i, Y_i \in \{0,1\}} \cdot \mathbf{1}_{W_i \in \mathbb{R}^p}$$

# To go further

X: the random variable representing the treatment group (1 for treatment), $X \in \{0, 1\}$

Y: the outcome variable, representing the number of VFDs in the ICU, $Y \in [\![0, 28]\!]$,

Z: the variable indicating whether the patient died (1 for death), $Z \in \{0, 1\}$,

- $W \in \mathbb{R}^p$: a vector of covariates (terms susceptible to interact with the main covariate X)

## Goals

### Treatment failure

#### VFDs

$$\begin{cases} Y \mid Z = 0, X = x, W = w \sim \mathcal{NB}(\mu(x, w), \theta), x \in \{0, 1\}, w \in \mathbb{R}^p \\ Y \mid Z = 1, X = x, W = w \sim \mathcal{B}(\pi(x, w)) \end{cases}$$

with:

$$\mu(x, w) = \exp\left(\beta_0 + \beta_1 x + \sum_{j=1}^{p} \gamma_j w_j + \sum_{j=1}^{p} \nu_j (x \odot w_j)\right),$$

$$\pi(x, w) = \frac{\exp\left(\alpha_0+\alpha_1 x+\sum_{j=1}^{p} \eta_j w_j+\sum_{j=1}^{p} \zeta_j (x\odot w_j)\right)}{1+\exp\left(\alpha_0+\alpha_1 x+\sum_{j=1}^{p} \eta_j w_j+\sum_{j=1}^{p} \zeta_j (x\odot w_j)\right)}.$$

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## Goals

### Treatment failure

### VFDs

Zero-inflated negative binomial model

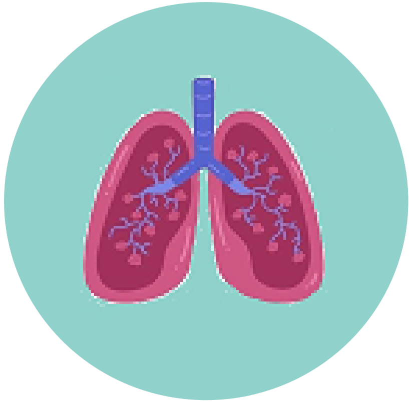
$$f_{Y|X,Z,W}(y \mid x, z, w) = \begin{cases} \left[\pi(x,w) + (1-\pi(x,w))\left(\frac{\theta}{\theta+\mu_{x,w}}\right)^{\theta}\right] \cdot \mathbf{1}_{\{\mu_{x,w}>0,\ \theta>0\}} \cdot \mathbf{1}_{x\in\{0,1\}} \cdot \mathbf{1}_{w\in\mathbb{R}^p}, \text{ if } y=0, \\ (1-\pi(x,w))\frac{\Gamma(y+\theta)}{y!\,\Gamma(\theta)}\left(\frac{\theta}{\theta+\mu_{x,w}}\right)^{\theta}\left(\frac{\mu_{x,w}}{\theta+\mu_{x,w}}\right)^{y} \cdot \mathbf{1}_{\mathbb{N}}(y) \cdot \mathbf{1}_{\{\mu_{x,w}>0,\ \theta>0\}} \cdot \mathbf{1}_{x\in\{0,1\}} \cdot \mathbf{1}_{w\in\mathbb{R}^p}, \ y>0 \end{cases} \quad (3.3.1)$$

The likelihood for i.i.d sample (Yn, Xn, Zn, Wn) drawn from (Y,X, Z, W):

$$L(\mu,\theta) = \begin{cases} \prod_{i=1}^{n} \pi(X_i,W_i) + (1-\pi(X_i,W_i))\left(\frac{\theta}{\theta+\mu_{X_i,W_i}}\right)^{\theta}, \text{ for all } i \text{ such that } Y_i = 0, \\ \prod_{i=1}^{n} (1-\pi(X_i,W_i))\frac{\Gamma(Y_i+\theta)}{Y_i!\,\Gamma(\theta)}\left(\frac{\theta}{\theta+\mu_{X_i,W_i}}\right)^{\theta}\left(\frac{\mu_{X_i,W_i}}{\theta+\mu_{X_i,W_i}}\right)^{Y_i} \text{,otherwise} \end{cases}$$

# Conclusion

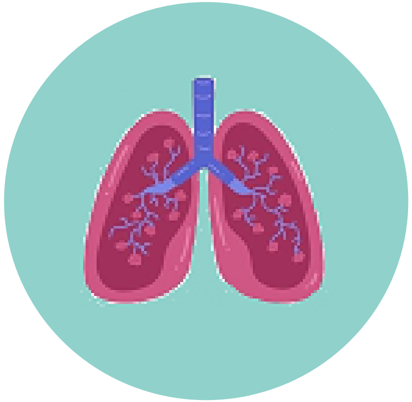
- We observed a reduced incidence of radiographic progression in patients who received methylprednisolone.
- Bayesian approach confirmed a significant probability a beneficial effect of the treatment, solely on the number of VFDs, while frequentist approach only confirmed an effect on mortality rate, which is though a component of ventilation-free days. **Therefore, a positive effect on ventilation-free days is likely.**



# Limits

- The main limitation of the use of this method in our case lies in the study design, which has been thought for a frequentist approach: in the regulatory setting, the design of a Bayesian clinical trial involves pre-specification of and agreement on both the prior information and the model.*

- Also, the main criticism towards VFDs is that a single composite risk estimate does not adequately distinguish between component risks and effect measures**



*** Nadir Yehya, Michael O. Harhay, Martha A. Q. Curley, David A. Schoenfeld, and Ron W. Reeder.Reappraisal of Ventilator-Free Days in Critical Care research. American Journal of Respiratoryand Critical Care Medicine, 200(7):663–664, 2019*

# THANK YOU